Interpretable Feature Context Effects

Linear Context Logit (LCL) and Decomposed Linear Context Logit (DLCL): how does the choice set change people's preferences?

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DA5453 · Learning from Human Preferences

Paper: Tomlinson & Benson, *Learning Interpretable Feature Context Effects in Discrete Choice*

22/10/26

Recap: Linear Context Logit (LCL)

$$\Pr(i, C) = \frac{\exp\left([\theta + A x_C]^\top x_i\right)}{\sum_{j \in C} \exp\left([\theta + A x_C]^\top x_j\right)}$$

$x_C = \frac{1}{|C|} \sum_{j \in C} x_j$

The mean features of the choice set

A_{pq}

How much feature q in the mean of the choice set effects feature p

LCL example: choosing a restaurant

Features: (price, speed, wine)

$\theta = [-1,\ 1,\ 1]$

$$A = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Choice set	mean price	$\theta + A x_C$
Quick Taco, Burger Express	-1	$[-1,\ 2,\ 0]$
Steak Deluxe, Burger Express	0	$[-1,\ 1,\ 1]$
Steak Deluxe, Château Caviar	$+1$	$[-1,\ 0,\ 2]$

$A_{21} = -1 \quad \rightarrow \quad$ pricier menu $\Rightarrow$ speed matters **less**

$A_{31} = +1 \quad \rightarrow \quad$ pricier menu $\Rightarrow$ wine matters **more**

Why LCL is not enough

Weight on feature p in set C:

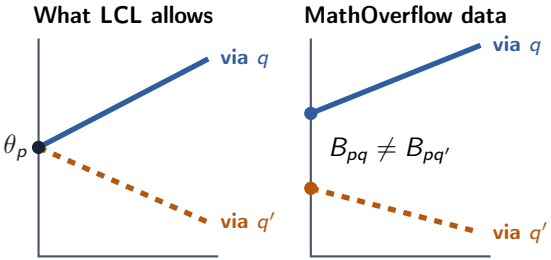
$$\underbrace{\theta_p}_{\text{one shared intercept}} + \sum_q A_{pq}\,(x_C)_q$$

One intercept for every effect

Effects from different features must start from the same baseline

One kind of chooser

Cannot model subpopulations whose choices follow different effects.



Mixed Logit

$$\Pr(i, C) = \sum_{m=1}^{M} \pi_m \frac{\exp\left(\theta_m^\top x_i\right)}{\sum_{j \in C} \exp\left(\theta_m^\top x_j\right)}$$

Why mixed logit?

- Models M groups, which have a share π_m in the population.
- Each group follows an MNL with θ_m.
- The overall probability does not follow IIA.

Decomposed Linear Context Logit (DLCL)

$$\Pr(i, C) = \sum_{k=1}^{d} \pi_k \frac{\exp\left([\, B_k + A_k\,(x_C)_k\,]^\top x_i\right)}{\sum_{j \in C} \exp\left([\, B_k + A_k\,(x_C)_k\,]^\top x_j\right)}$$

- A mixed logit with k groups – models one feature creating an MNL model due to its context effect
- B_k: Weights when feature k is 0
- A_k: How the weights change as the feature k changes
- $2d^2 + d$ parameters, as opposed to $d^2 + d$ parameters in LCL.

DLCL example: two effects, two intercepts

Weight on feature 3 in four choice sets

Set	x_C	observed	LCL forced
C_1	$(1,0,0)$	1	1
C_2	$(2,0,0)$	2	2
C_3	$(0,1,0)$	3	3
C_4	$(0,2,0)$	4	**6**

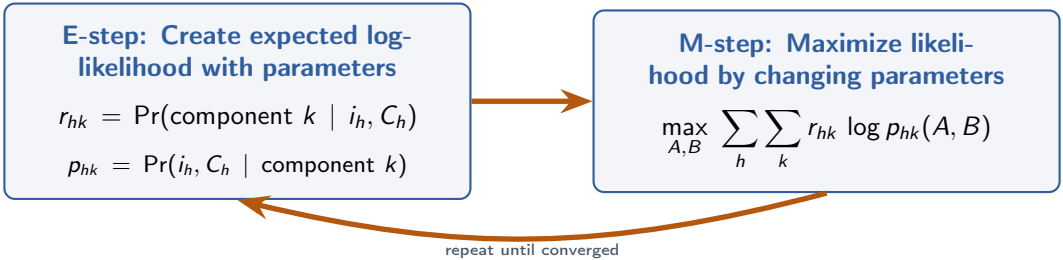
LCL: one intercept θ_3

$C_1, C_2 \Rightarrow \theta_3 = 0,\ A_{31} = 1;$
$C_3 \Rightarrow A_{32} = 3;$
$\Longrightarrow$ C_4 must be 6.

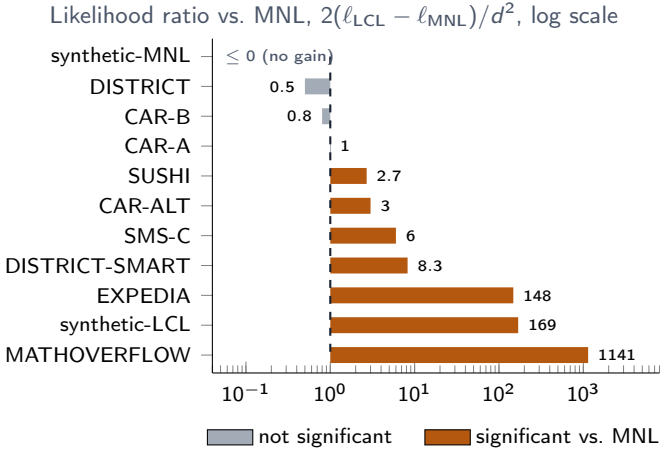
DLCL: one line per feature

Feature 1: $B_{31} + A_{31}t = 0 + t$
Feature 2: $B_{32} + A_{32}t = 2 + t$

Expectation–Maximization (EM) for DLCL

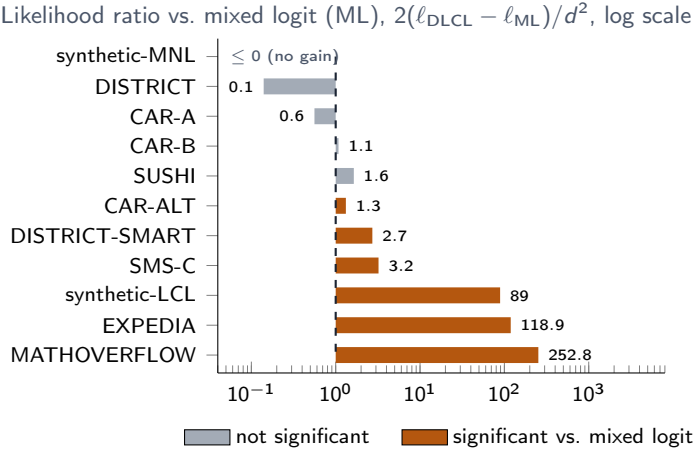


How good is the fit of LCL?



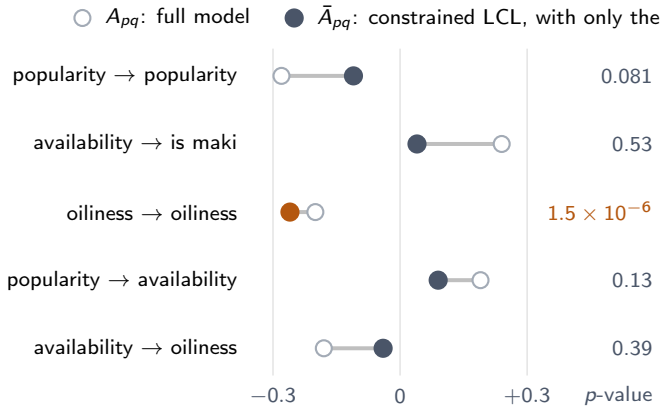
significant: $p < 0.001$ dashed line at 1 = what pure noise would give

How good is the fit of DLCL?



significant: $p < 0.001$ dashed line at 1 = what pure noise would give

SUSHI



Test effects one at a time

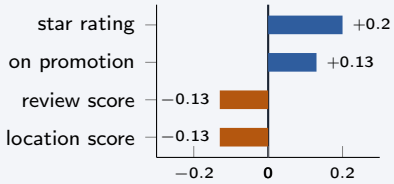
Many entries drop when fit alone.

Oily menus deter oiliness

More oily sushi options reduce oily sushi demand.

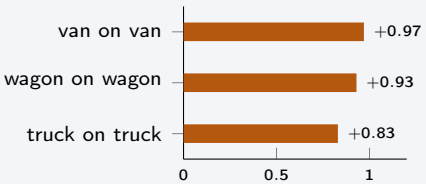
When choice sets are not random

EXPEDIA: hotel search



Effect of each feature on the price weight – positive is less price averse.

CAR-ALT: vehicle survey



There is a large positive self-effect, but surveys offered options which were most likely to be purchased.

Thank you

Questions?